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QoQ-Med3: A Multimodal Reasoning Foundation Model for Clinical Analysis

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ABSTRACT

Multimodal reasoning-based foundation models (MRFMs) hold considerable promise for addressing key challenges in medical practice, yet their readiness for real-world deployment remains insufficiently explored. To bridge this gap, we developed two MRFMs (QoQ-Med3 and QoQ-Med3-MIMIC) and systematically evaluated their (i) generalizability to previously unseen clinical modalities and tasks, (ii) transferability to held-out datasets collected across different clinical sites, and (iii) robustness to real-world challenges like cross-site heterogeneity. Our results demonstrate that these models can learn transferable representations across modalities, tasks, and heterogeneous clinical datasets. QoQ-Med3 achieves an overall balanced accuracy of 71.3% and a F1 of 0.349, superceding all open source and closed source models including GPT-4o, with particularly pronounced gains in understudied modalities such as ultrasound and mammography. The model trained on public clinical data only generalized to the both the held-out MIMIC-IV and the private JHU PMAP dataset collected at Johns Hopkins University hospital. In addition, the extrinsic hallucination rates are reduced by 44.4 percent after training. Collectively, our findings highlight both the potential of multimodal reasoning-based clinical foundation models and the critical next steps required to make them robust and reliable for real-world deployment.

Introduction

AI for clinical diagnosis has evolved significantly over the past decade, with numerous computational models developed to assist clinicians in organizing patient records^{1,2}, formulating diagnoses^{3,4}, interpreting clinical images^{5,6}, and other clinical tasks⁷. These advancements have substantially improved healthcare efficiency and accuracy across multiple specialties^{8,9}. Recently, the rise of foundation models has enabled quick adaptation of pre-trained models in the radiology¹⁰, psychiatry¹¹, and ophthalmology¹² domains. However, these methods still suffer from two main limitations. Firstly, clinical data spans multiple modalities across 1D (ECG, EEG), 2D (Chest X-ray, dermoscopy, mammography), and 3D (CT Scans, MRI). While existing models like BiomedGPT¹³ and MedFlamingo¹⁴ are including vision, no existing model has been able to integrate both 1D sensor data with 2D/3D images. The heterogeneity across specialties and modalities^{15,16} often leads to settings where modalities compete rather than synergize, leading to 3 performance¹⁷⁻²⁰. This necessitates careful retraining or fine-tuning strategies to balance heterogeneous distributions while enriching these models with clinical knowledge.

Secondly, despite the proliferation of foundation models for EHR, they remain predictive “black box” models that generate a single, definitive answer without revealing their underlying analytical process²¹⁻²³. This significantly impedes their practical adoption in clinical settings, as healthcare professionals might hesitate to trust diagnostic suggestions without understanding the reasoning that produced them²⁴. Transparency in the decision-making process is not merely a preference but a necessary component for responsible clinical implementation, regulatory compliance, and effective human-AI collaboration in healthcare environments²⁵⁻²⁷. This problem is exacerbated since models often fail silently in the face of heterogeneous temporality^{28,29}, long context windows³⁰, and cross-site heterogeneity in clinical workflows and coding practices³¹ within non-stationary healthcare environments that change data distributions both across time and population^{32,33}. Recent benchmarking efforts have begun to address these gaps by evaluating fairness and interpretability across diverse populations³⁴ and proposing clinically

meaningful evaluation suites beyond ICU-restricted cohorts³⁵, yet most models remain trained on narrowly-scoped datasets that provide limited insight into real-world utility³⁰.

To tackle these two challenges of heterogeneity and interpretability, we leverage recent advances in multimodal reasoning foundation models (MRFMs) integrating diverse clinical modalities including imaging, time series, and textual records^{13,36–40}. The representations learnt by multimodal foundation models have proven effective for a variety of downstream healthcare-related data and tasks, including leveraging MRFMs to analyze highly heterogeneous Electronic Health Records (EHRs) and radiology data^{30,35,41,42}. Secondly, GPT-5⁴³ and Deepseek R1⁴⁴ have inspired efforts to create specialized clinical reasoning systems^{45–47} capable of answering complex clinical questions and generating comprehensive clinical reports^{13,36}. Reasoning allows models to think explicitly in a more logical and systematic way with evidence from the inputs and their own knowledge⁴⁸, all of which are essential for robust clinical diagnosis and doctor-in-the-loop usage^{49,50}.

Wornow *et al.* found that most models are trained on small, narrowly-scoped clinical datasets (e.g., MIMIC-III) or broad, public biomedical corpora (e.g., PubMed) and are evaluated on tasks that do not provide meaningful insights on their usefulness to health systems and further released a 141M parameter clinical foundation model pretrained on the structured EHR data of 2.57M patients³⁰. Yu *et al.* presented a comprehensive benchmark that evaluates the performance, fairness, and interpretability of foundation models, both as unimodal encoders and as multimodal learners, using the publicly available MIMIC-IV database³⁴. Zhu *et al.* trained a foundation model to predict the risk of Alzheimer’s disease using EHRs⁵¹. Sivarajkumar *et al.* introduced a multimodal foundation model that natively encodes structured EHR time-series via a CNN-Transformer encoder and fuses it with unstructured EHRs through cross-modal attention into a LLaMA-based decoder.⁴⁰ However, the biggest model size till date are no greater than 20B parameter. In contrast, our model is currently the largest open clinical MLLM in the field at a size of 32B. It is also the only model till date that can take in 1D, 2D (Chest X-ray, mammography, dermoscopy, histopathology) and 3D (ultrasound, MRI and CT scan) inputs. Moreover, by leveraging a stronger backbone, the different modalities are better aligned which thus allows it to learn better representation across the different heterogeneous modalities, thus setting it apart from previous medical foundation models.

Renc *et al.* combined a system with foundation models to compute dynamic, personalized risk probabilities for clinician-defined critical events using MIMIC-IV⁴². Guo *et al.* attempted to improve the temporal robustness of EHR models via pre-training³². Pange *et al.* proposed a suite of clinically meaningful tasks spanning patient outcomes, early prediction of acute and chronic conditions, including desiderata for robust evaluations using EHR data consisting of 5 million patients from Columbia University Irving Medical Center (CUMC)³⁵. Compared to existing works, our model is the only model that takes in time series data and output its thinking process, along with the bounding box annotation highlighting the salient region made during the thinking process which are deemed essential for real-world deployment^{52–54}. As a result of our stronger backbone and novel architecture choices, our model is better able to produce step-by-step explanations that link clinical knowledge to specific image features and patient data. This makes its predictions easier to trace and audit, as intermediate reasoning steps reveal how different modalities are combined. In contrast to earlier medical foundation models that often rely on implicit representations or shallow fusion, our models thus provide more transparent and structured justifications, allowing clinicians to better assess the validity of its conclusions and identify potential failure modes.

By marrying the two ideas, of multimodal fusion and cross-modality reasoning, we introduce a new set of multimodal clinical reasoning models that integrates visual, time series, and textual data for comprehensive analysis of clinical records and are capable of facilitating more holistic diagnostic reasoning^{37,55}. As an example of our model’s capabilities in Fig 1, we see that such models also advance model interpretability whilst allowing clinicians to assess the model’s diagnosis with ease. Building upon Domain-aware Relative Policy Optimization (DRPO), a reinforcement learning algorithm to enable reasoning over highly heterogeneous multimodal data, we present QoQ-Med3, a MRFM building on top of Qwen3-VL that sets state-of-the-art clinical diagnostic performance across 7 clinical modalities, including ECG, CT Scan, Chest X-Ray, Fundus, Ultrasound, Mammography and Pathology. Upon QoQ-Med3, we aim to investigate: **“how can large-scale MRFMs be effectively leveraged to improve performance and generalizability across diverse, multimodal downstream clinical tasks across real-world conditions and out-of-distribution settings?”** We do so by answering the following research questions (RQs):

1. **RQ 1: Do MRFMs pretrained on different datasets reflect increased generalization and robustness to new clinical modalities and tasks?** To answer this, we further design, propose and conduct experiments using new models, QoQ-Med3 and QoQ-Med3-MIMIC and publicly release our model, training pipeline, and reasoning traces generated by the model across 2.61 million question-answer pairs at [this link](#). This marks one of the largest resources for transparent and reproducible multimodal reasoning in the clinical domain. We further conduct modality ablation and real-world clinician analysis to understand how the different modalities contribute to the eventual reasoning-based clinical outcome. Our findings show that QoQ-Med3 trained with DRPO achieves an overall balanced accuracy of 71.3% and a F1 of 0.349, a 14.7 percentage point improvement over the strongest baseline, with particularly pronounced gains in understudied modalities such as ultrasound and mammography. Clinician annotations further confirm that the majority of reasoning

Table 1. Summary of data demographics. Length of stay and survival percentage for the JHU and MIMIC-IV datasets are reported across the hospital admissions, while the remaining demographic makeup is reported across the patients. MIMIC-IV tasks include diagnosis prediction (open-ended and multiple choice), 48-hour survival likelihood, and length of stay prediction using tabular EHR data, radiology images, and clinical notes. JHU tasks include length of stay, ventilator duration, sepsis diagnosis, bleeding severity, 30-day mortality, and in-hospital mortality prediction using tabular EHR data and ECG reports.

	MIMIC-IV	JHU PMAP
Patients (<i>n</i>)	218,982	845,904
Hospital stays (<i>n</i>)	537,532	8,323,553
Age (mean $\pm$ std.)	55.9 $\pm$ 20.1	52.4 $\pm$ 22.6
Gender		
Male	47.2 %	43.1 %
Female	52.8 %	56.8 %
Unknown/other	0.0 %	0.1 %
Race and ethnicity		
White	66.4 %	59.5 %
Black	16.3 %	28.3 %
Asian	3.6 %	2.6 %
Hispanic	5.9 %	4.8 %
Other/unknown	7.7 %	4.7 %
Length of stay in hours (mean [IQR])	114.9 [27.7, 135.5]	8.7 [2.1, 13.8]
Survival percentage	97.7 %	98.6 %

steps correspond to diagnostically relevant evidence.

- RQ 2: Are the findings from foundation models pre-trained on non-EHR data generalizable to and capable of improving medical diagnosis on EHR data collected from different clinical sites and settings?** To answer this RQ, we investigate how the performance of a medical foundation model pre-trained on CLIMB, a non-EHR collection of medical datasets, transfers to MIMIC-IV, a widely used benchmark for EHR-based clinical prediction. Further, we systematically compare model performance across the MIMIC-IV and Johns Hopkins University (JHU) datasets, examining scalability and generalizability across institutions, clinical specialties, and care settings. Our results show that QoQ-Med3 pretrained on non-EHR data (CLIMB) achieves 3.4-fold higher BioBERT similarity and 8.1% higher accuracy than the base model on MIMIC-IV in a zero-shot setting, with further gains when combined with MIMIC-IV finetuning. These improvements generalize to the held-out JHU dataset, as models trained on MIMIC or CLIMB exhibits superior performance on JHU dataset as well.
- RQ 3: How robust are medical foundation models to failure modes such as model hallucinations, and how well do they perform when evaluated on outcomes beyond prediction?** To address this question, we conduct experimental analyses to examine model robustness, assess potential biases across diverse patient populations through subgroup analyses, and systematically evaluate the prevalence and characteristics of model hallucinations. In particular, we curate an exhaustive criteria to judge incorrect model outputs in terms of intrinsic and extrinsic hallucinations, and identify substantial concordance amongst two human annotators and one LLM-as-a-judge (Fleiss' $\kappa = 0.792$). We see that the number of extrinsic hallucinations reduce by 44.4% after training on CLIMB and MIMIC datasets, thereby enforcing that domain-specific training strongly improves model reasoning performance.

Results

Cohort Characteristics and Outcome Prevalence

The demographic breakdown of both the MIMIC-IV and JHU datasets are presented in Table 1. From Table 1, we see that the demographic distributions are similar in both datasets, particularly across age, gender, aggregated race, and survival percentage. The length of stay is larger by orders of magnitude for MIMIC-IV since it includes hospital admissions in the intensive care unit and emergency department. Demographic information for aggregated race, gender, and insurance type in MIMIC-IV data is visualized in Fig. 5(b) through (d).

Table 2. Performance comparison with standard deviations. Acc: Accuracy, F1: F1 Score, CXR: Chest X-ray.

QoQ-Med3 outperforms various other RL training methods and SFT across 7 out of 8 medical imaging domains in F1 score. The metrics are averaged across four separate training runs. All methods in Table 2 share the same Qwen3-VL-8B backbone and reward design. P-values are computed using Welch’s t-test across four independent training runs. “vs baseline” compares QoQ-Med3 against the best-performing baseline in each column; “Med3 vs Med” compares against the previous generation model.

Model	CXR		Mammo.		Dermoscopy		CT Scan	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1
SFT	.688±.030	.078±.013	.481±.015	.056±.004	.640±.029	.158±.000	.525±.000	.236±.000
ReMax ⁵⁶	.636±.049	.120±.030	.577±.063	.033±.020	.644±.009	.257±.018	.567±.026	.228±.001
RE++ ⁵⁷	.730±.027	.082±.032	.660±.087	.076±.011	.635±.008	.237±.079	.529±.013	.247±.039
RLOO ⁵⁸	.752±.015	.086±.003	.471±.002	.068±.006	.636±.008	.216±.028	.534±.014	.224±.042
GRPO ⁵⁹	.703±.012	.095±.007	.466±.000	.059±.000	.646±.003	.244±.016	.524±.002	.236±.001
QoQ-Med ³⁷	.687±.005	.115±.014	.756±.008	.253±.006	.715±.006	.407±.006	.570±.012	.309±.013
QoQ-Med3	.750±.001	.242±.027	.759±.012	.241±.029	.693±.009	.350±.009	.567±.005	.254±.003
<i>p-value (vs baseline)</i>	.807	<.001	.106	<.001	<.001	<.001	.843	.044
<i>p-value (Med3 vs Med)</i>	<.001	<.001	.693	.472	.008	<.001	.668	.002

Model	Fundus		Ultrasound		MRI		Pathology		Overall	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
SFT	.715±.051	.066±.010	.548±.028	.235±.000	.567±.038	.197±.000	.652±.000	.083±.000	.602±.014	.139±.001
ReMax ⁵⁶	.678±.015	.089±.006	.547±.044	.147±.008	.547±.012	.264±.018	.706±.007	.270±.011	.596±.002	.176±.001
RE++ ⁵⁷	.672±.003	.098±.013	.519±.010	.136±.012	.651±.022	.420±.025	.668±.024	.254±.026	.621±.020	.202±.053
RLOO ⁵⁸	.670±.003	.099±.013	.519±.012	.144±.014	.658±.032	.432±.020	.699±.011	.216±.024	.611±.002	.189±.003
GRPO ⁵⁹	.670±.001	.086±.002	.520±.009	.146±.011	.631±.027	.395±.030	.715±.004	.286±.010	.609±.005	.193±.003
QoQ-Med ³⁷	.672±.004	.093±.008	.555±.014	.223±.021	.789±.019	.625±.028	.708±.010	.265±.016	.666±.009	.295±.013
QoQ-Med3	.738±.001	.310±.022	.660±.023	.375±.031	.892±.011	.827±.017	.809±.038	.550±.089	.713±.000	.349±.007
<i>p-value (vs baseline)</i>	.433	<.001	<.001	.002	<.001	<.001	.015	.009	.002	.010
<i>p-value (Med3 vs Med)</i>	<.001	<.001	<.001	<.001	<.001	<.001	.010	.006	.001	.001

Performance on CLIMB

CLIMB, introduced by Dai et al.⁵⁵, is a large scale clinical diagnosis dataset across 13 modalities. In the training of QoQ-Med3, we utilize 8 modalities, including including ECG, CT Scan, Chest X-Ray, Fundus, Ultrasound, Mammography, Pathology and MRI. Table 2 presents a comprehensive comparison of QoQ-Med and QoQ-Med3, our latest model, against several critic-free RL training methods across eight medical imaging modalities. Baselines include ReMax⁵⁶, Reinforce++⁵⁷, RLOO⁵⁸, GRPO⁵⁹, which are popular MRFM post-training methods that are known to elicit model’s reasoning capabilities. QoQ-Med3, trained with DRPO, is a novel training method that is optimized specifically for highly heterogeneous multimodal data. We note that the overall F1 is macro-averaged across all classes including rare conditions with very few positive samples, which produces lower absolute values than micro-averaged or binary metrics. In this setting, relative improvements are more informative. QoQ-Med3 achieves a 72.8% relative F1 improvement over the strongest baseline (0.349 vs. 0.202 for RE++). Domains with fewer classes, such as MRI, yield substantially higher absolute F1 scores. Understudied modalities as defined in Method Section also benefit substantially. Ultrasound F1 improves from 0.235 to 0.375, and mammography F1 rises from 0.076 to 0.241.

Comparing QoQ-Med3 to our previous generation model QoQ-Med reveals further improvements in most domains. As illustrated in Fig. 2(a), the improvements over GRPO are most pronounced in the understudied modalities. Overall accuracy increases from 0.666 to 0.713, and F1 improves from 0.295 to 0.349, both statistically significant at $p = .001$. The largest generational gains occur in fundus F1, which more than triples from 0.093 to 0.310, and in pathology F1, which doubles from 0.265 to 0.550. MRI performance continues to improve with F1 rising from 0.625 to 0.827. However, QoQ-Med3 does show slight regressions in dermoscopy, CT and mammography F1. These regressions suggest potential trade-offs in the training process that merit further investigation, though the overall performance strongly favors QoQ-Med3.

As shown in Fig. 2(b), our 32B variant, QoQ-Med-32B achieves the best performance across all clinical domains as

compared to current open-source MLLMs. Compared to the closed-source commercial models, it achieves the best performance against GPT-4o⁶⁰, while surpassing the reasoning model GPT-o4-mini⁶¹ in all domains except MRI.

Performance on MIMIC-IV

Figure 3 presents the performance comparison on the MIMIC-IV benchmark across five question types. For open-ended diagnosis tasks (T1, T2), we use semantic similarity based on BioBERT embeddings of ICD codes, following Kim et al⁶². Predicted and ground-truth ICD codes are embedded using BioBERT, and performance is measured via cosine similarity in the embedding space (Figure 3a-b). T3 through T5 are multiple choice questions covering primary diagnosis (T3), 48-hour survival prediction (T4), and length-of-stay prediction (T5), evaluated by accuracy and F1 score (Figure 3c-h). Overall metrics are shown in Figure 3(i)-(k). Detailed task descriptions are provided in Dataset Section. We observe three key findings.

- First, **domain-specific training on MIMIC substantially improves performance**. Qwen3-MIMIC achieves a nearly five-fold improvement in BioBERT similarity (0.182 vs. 0.038), as shown in Figure 3(k) and a 16.8% increase in overall accuracy compared to the base Qwen3-VL model in Figure 3(j).
- Second, **QoQ-Med trained exclusively on CLIMB for general clinical diagnosis demonstrates strong transferability to MIMIC-IV** without any exposure to this dataset during training. QoQ-Med-ZeroShot outperforms the base Qwen3-VL model across all overall metrics, achieving 3.4-fold higher BioBERT similarity and 8.1% higher overall accuracy.
- Third, the **combination of CLIMB pretraining followed by MIMIC-IV finetuning** (QoQ-Med-MIMIC) yields the best overall performance, achieving the highest BioBERT similarity of 0.199, competitive accuracy of 0.858, and the highest F1 score of 0.466.

Modality Ablations

We tested how the model integrates multiple modalities and how much each modality contributes to the final diagnostic accuracy via MIMIC-IV⁶³ dataset. In this experiment, we filter the test dataset so that the sample only contains data with all three modalities, including time series (ECG), vision (X-ray) and text (EHR). As shown in Figure 6, we found taking full inputs across ECG, chest X-ray images, and electronic health records (EHR) gives better performance than any ablation of these modalities, signaling that QoQ-Med3 is able to effectively aggregate information across all modalities.

Performance on JHU PMAP

Figure 4 presents the results on the Johns Hopkins Hospital dataset. We evaluate model-generated responses using four variants: Qwen3-VL-8B, Qwen3-8B-MIMIC, QoQ-Med3-ZeroShot, and QoQ-Med3-MIMIC. We designed six multiple-choice question types, as defined in Dataset Section that cover essential clinical questions arising in emergency room and inpatient settings. For each question type, we randomly sample 10,000 complete patient records containing the relevant features from the database, generate model responses, and compute accuracy and F1 score as evaluation metrics. We make a best effort to balance the ground-truth answer distribution across the sampled population; however, some tasks remain inherently imbalanced: for example, ventilator duration, as most patients do not require ventilation. Overall, the base QoQ-Med3 model clearly outperforms the base Qwen3-VL model. Fine-tuning on the MIMIC-IV dataset further improves performance on clinical domain knowledge, with both Qwen3-MIMIC and QoQ-Med3-MIMIC achieving higher scores. Notably, we observe consistent gains in both accuracy and F1 score on the length-of-stay (LOS), 30-day mortality, and in-hospital mortality tasks. These tasks are generally more challenging due to their relatively uniform answer distributions.

Failure Mode Analysis

In Figure 5(a), incorrect model outputs for multiple choice tasks are categorized into intrinsic and extrinsic hallucinations for each model variant using an LLM-as-a-judge ($n = 1539$ multiple choice validation samples for each model variant). Definitions are included in the Method Section and define a taxonomy of incorrect model predictions; the count and unconditional rate of hallucination across all samples is displayed in Table 3.

A χ^2 -test of independence was performed to examine the relationship between model variant and hallucination type for each pair of model variants. Table 4 reports the resulting test statistics, comparing the per-sample rate of hallucinations. All pairwise comparisons reported statistically significant differences except for two pairs: Qwen3 and QoQ-Med3-ZeroShot ($p = 0.0689$), and Qwen3-MIMIC and QoQ-Med3-MIMIC ($p = 0.808$). In particular, the percentage of extrinsic and intrinsic hallucinations reduced by 44.4% and 6.16% respectively after training on CLIMB and MIMIC datasets; explicitly, we compare the rate of hallucinations from QoQ-Med3-MIMIC to the Qwen3 baseline. In addition to LLM-as-a-judge, two human annotators independently labeled a random sample of data ($n = 100$), comprised of data from all four model variants in shuffled order. Inter-annotator agreement across two human annotators and one LLM-as-a-judge was substantial (Fleiss' $\kappa = 0.792$). Examples

Table 3. Breakdown of hallucination failure modes by model. Hallucination categories are defined in Method Section. Categorization was performed by GPT-5.5, which received as input a detailed system prompt explaining the judging task, the text of the formatted Q&A pair, the ground truth, and model output. Each model had a total of $n = 1539$ samples categorized.

Model	Count			Unconditional Rate	
	Extrinsic	Intrinsic	Total Incorrect	Extrinsic	Intrinsic
Qwen3	537	225	762	34.9%	14.6%
Qwen3-MIMIC	309	217	526	20.1%	14.1%
QoQ-Med3-Zeroshot	490	208	698	31.8%	13.5%
QoQ-Med3-MIMIC	298	211	509	19.4%	13.7%

Table 4. Pairwise chi-square tests of independence between model predictions. Each row reports the chi-square statistic (χ^2), degrees of freedom (df), sample size (N), and p-value for a pairwise comparison between two models.

Model 1	Model 2	χ^2	df	N	p-value
Qwen3	Qwen3-MIMIC	92.707	2	3078	7.40×10^{-21}
Qwen3	QoQ-Med3-Zeroshot	5.350	2	3078	6.89×10^{-2}
Qwen3	QoQ-Med3-MIMIC	104.281	2	3078	2.27×10^{-23}
Qwen3-MIMIC	QoQ-Med3-Zeroshot	57.150	2	3078	3.89×10^{-13}
Qwen3-MIMIC	QoQ-Med3-MIMIC	0.425	2	3078	8.08×10^{-1}
QoQ-Med3-Zeroshot	QoQ-Med3-MIMIC	65.895	2	3078	4.91×10^{-15}

of labeled hallucinations with reasoning traces produced by the LLM-as-a-judge are included in Supplement, as well as each pairwise Cohen’s kappa between annotators.

We examined when QoQ-Med3-MIMIC hallucinated on MIMIC-IV data in order to investigate its performance on demographic disparities in a clinical context. Considering that we infer only from the hallucination analysis completed by the LLM-as-a-judge, these observations should not be interpreted as evidence of systematic bias in model behavior, but rather potential insight and mere exploration into an unintentional model skew.

The model was not provided with race or insurance information at any stage of training or evaluation. By excluding these features, we were able to examine demographic differences in model performance independent of explicit demographic conditioning. In Figures 5(b) and (e), hallucination rates are higher for less prevalent racial groups in the training data. Asian patients show hallucination rates 14.5 percentage points above those of White patients, while the remaining aggregated racial groups exceed the White rate by 1.76 percentage points on average. This pattern suggests that under-representation in the training distribution may be associated with elevated hallucination risk. A similar, but weaker, trend is observed for insurance status in Figures 5(d) and (g), where less prevalent insurance categories – Medicaid and Other – exhibit hallucination rates comparable to those of more common insurance types, despite their lower representation in the training data.

In contrast, gender, which is an attribute explicitly provided as part of a patient’s history to the model, does not exhibit this disparity. As shown in Figures 5(c) and (f), hallucination rates for Female and Male patients closely mirror their respective prevalence in the training dataset. This alignment indicates no measurable gender-specific amplification of hallucinations and suggests that explicitly modeled demographic attributes may be less susceptible to representation-driven hallucination effects.

Each MIMIC-IV Q&A pair is attributed to a primary diagnosis category by identifying the patient’s primary reason for hospital admission, as identified by the diagnosis labeled with the lowest sequence number for each admission. Figures 7 and 8 showcase an inverse relationship between the frequency of primary diagnosis categories in the MIMIC-IV training dataset and the hallucination rate of Qwen3, QoQ-Med3-Zeroshot, and QoQ-Med3-MIMIC. This behavior may suggest that the model can reduce its likelihood of hallucinating across medical domains if it sees more cases of each disease group in its training pipeline.

Discussion

As in RQ 1, we explored the question “RQ 1: Do multimodal foundation models pretrained on different datasets reflect increased generalization and robustness to new clinical modalities and tasks?” Our results demonstrate that large-scale multimodal clinical pretraining provides a strong foundation that enhances downstream task performance after domain-specific finetuning. Specifically, we found vision and text contribute more to the final accuracy and F1 scores than ECG. The model also demonstrates sophisticated understanding of imaging physics. In the brain tumor cases in Fig. 1(d), the model correctly discusses T1 and T2 signal characteristics, noting that tumors often appear “hyperintense on T2” due to associated edema and

altered tissue properties. It identifies the tumor as “round, well-circumscribed” with a “clear boundary” from surrounding tissue and “clearly distinct from the surrounding brain tissue”, which are all clinically relevant observations that support the diagnosis. Notably, in the negative case shown in Supplement, the model appropriately considers meningioma characteristics but ultimately reaches the correct “No Tumor” conclusion despite some reasoning about tumor features, demonstrating appropriate calibration.

To evaluate the clinical utility and real-world readiness of QoQ-Med3’s reasoning process, we collaborated with clinicians to annotate reasoning traces on randomly selected validation samples, with phrases marked by relevance to diagnosis: high (pink), low (yellow), or none (aqua). In general, we see around 45.3% of reasoning traces rated as highly relevant, and 100% of the rated samples contain highly relevant reasoning traces. Fig. 1 presents representative examples. Clinician review confirmed that the majority of reasoning steps corresponded to diagnostically relevant evidence with minimal unsupported content.

While the clinician annotation study presented here provides initial evidence of clinical utility, we acknowledge that this represents a retrospective evaluation on held-out validation data. The generalization to the held-out JHU PMAP dataset, collected independently at Johns Hopkins University across 845,904 patients and 8.3 million encounters, provides evidence of cross-site robustness that is a necessary precondition for prospective evaluation. Our hallucination analysis further establishes the safety characterization needed before clinical deployment, showing a 78.3% reduction in extrinsic hallucinations after training. We view the current work as establishing the technical foundation, with prospective clinician-in-the-loop evaluation as the natural next step.

Fig. 1(e) further highlights the model’s ability to combine multimodal inputs for complex clinical predictions. To estimate hospital length of stay, it integrates chest X-ray findings (“no obvious acute pathology”), ECG interpretation (“unremarkable. . . no signs of acute heart failure”), and laboratory values (“normal. . . no life-threatening abnormalities”) to predict a short stay. This kind of holistic reasoning across modalities is beyond the capability of unimodal models.

As in RQ 2, we explored the question “RQ 2: Are the findings from foundation models pre-trained on non-EHR data generalizable to and capable of improving medical diagnosis on EHR data collected from different clinical sites and setting?” Our findings suggest that clinical knowledge acquired through large-scale multimodal pretraining on non-EHR data CLIMB, which encompasses diverse diagnostic reasoning tasks, transfers effectively to downstream EHR data collected from different clinical sites and setting across several clinical prediction tasks including diagnosis, survival prediction, and length-of-stay estimation. Importantly, the strong zero-shot transfer results suggest that the model does not merely memorize dataset-specific patterns but acquires generalizable clinical reasoning capabilities. QoQ-Med3-ZeroShot, having never seen any MIMIC-IV data during training, still substantially outperforms the base model across all MIMIC-IV task types (Figure 3). This indicates that the CLIMB training corpus, while large, serves primarily to teach transferable multimodal clinical reasoning.

We hypothesize that reasoning-enabled models perform better because they can explicitly recall and apply domain-specific clinical knowledge during inference. For instance, in Fig. 1(a), the model recalls that hemorrhage appears differently by age and type—hypodense (dark) regions suggest acute bleeding, while hyperdense (bright) areas indicate older clots. It then examines the scan and notes that “the tissue appears uniformly gray, without any obvious bright or dark areas,” correctly predicting no hemorrhage. This step-by-step link from general knowledge to image-specific evidence provides an interpretable explanation. Similarly, in Fig. 1(b), the model applies dermoscopic criteria to distinguish diagnoses. It describes the lesion as “slightly raised” and “pinkish,” with a “rough, scaly surface,” consistent with sun-damaged skin in a sun-exposed area. It then systematically evaluates alternatives—melanoma, nevus, BCC, and AKIEC—before selecting BCC based on the accumulated evidence. This structured reasoning mirrors clinical decision-making and makes the model’s conclusions more transparent.

The full trace in Supplement reveals the model systematically considers each differential diagnosis—melanoma, nevus, BCC, and AKIEC, before converging on BCC based on the cumulative evidence. This structured elimination process mirrors clinical decision-making and further provides transparency into how the model weighs competing hypotheses.

As in RQ 3, we explored the question “how robust are medical foundation models to failure modes such as model hallucinations, and how well do they perform when evaluated on outcomes beyond prediction?” Although the model usually produces clinically relevant observations, it is not always robust and can hallucinate. It sometimes mistakes a past admission for the current admission, follows a faulty reasoning trace, or fails to integrate its findings into the final prediction. For instance, in Fig. 1(c), the model correctly notes that “the x-ray shows a pacemaker, which is a medical device,” and even recognizes this as a normal finding. However, it still predicts “No Finding” instead of “Support Devices,” indicating a disconnect between reasoning and classification. The full trace in Supplement shows further issues: the model hallucinates findings such as “increased lung density” and “pleural effusion,” and spends considerable effort on irrelevant details like patient positioning and radiographic markers. This example shows that identifying individual features does not ensure correct diagnosis if they are not properly weighted and combined.

In addition, the model struggles with fine-grained distinctions between similar conditions. For example, in lung ultrasound cases, it predicts “pneumonia” when the ground truth is “COVID-19.” While incorrect, its reasoning identifies relevant features—“hyperechoic areas,” “consolidation,” and patterns typical of pneumonia. Because COVID-19 and bacterial pneumonia often share ultrasound findings, distinguishing them usually requires clinical context (e.g., exposure history or lab markers)

not available in the image. This suggests that some errors reflect inherent diagnostic ambiguity rather than flawed reasoning. Such nuances are not captured by standard accuracy metrics but become visible through reasoning traces. Prospective clinician-in-the-loop evaluation is needed to assess real-world safety and performance under distribution shift.

As highlighted in our hallucination analysis, extrinsic hallucinations remain the principal bottleneck for reliable clinical deployment and real world readiness. While this suggests limitations of the current models, the judge model itself may also exhibit inaccuracies – in this experiment, we used GPT-5.5. The judge model may wrongly claim some part of the reasoning process is “hallucinated” due to its own limited understanding of clinical contexts. Additionally, our two human annotators comprise a small sample size of human review.

Finally, all judges in this analysis viewed only text data for rapid evaluation, which may skew the hallucination judgment. Future works should explore annotating the hallucination with clinical experts, stronger models, and multimodal input, in order to provide more accurate classification evaluation before real world deployment.

While QoQ-Med3 represents a first step towards multimodal reasoning models across vision and time series, future works could explore better architecture, data, or training methods that better balances the power of each modalities. The annotated traces also reveal patterns of low-relevance content generation. Across multiple examples, the model produces introductory statements (“To analyze the image, let’s go through the reasoning process step by step”) and generic explanations of imaging modalities that, while technically accurate, do not contribute to the specific diagnosis. We emphasize that clinician annotations consistently mark these segments in yellow (low relevance), suggesting opportunities for training refinements that encourage more focused, diagnostically pertinent reasoning. In particular, future work could explore how to best annotate reasoning traces at scale and then use the resulting data to guide the model toward reasoning most efficiently, for instance by penalizing generic introductory statements so that the reasoning concentrates on diagnostically pertinent evidence.

A further design consideration concerns our use of a frozen ECG-JEPA encoder for the time-series modality. Keeping the encoder frozen ensures a robust and stable representation of the time-series input, as no additional gradient updates are applied that could degrade or destabilize the features learned during large-scale ECG pretraining. The trade-off is that the time-series representation is not explicitly aligned with the downstream diagnostic objective: unlike the vision and text pathways, the 1D features are not natively learned during multimodal training and may therefore be less tailored to the final clinical understanding than if they were optimized jointly with the rest of the model. This reflects a tension between representational robustness and task-specific alignment. Future work should explore jointly training the time-series encoder together with the main model, for instance through gradual unfreezing or low learning-rate fine-tuning, to better align the 1D representation with downstream reasoning while preserving the robustness afforded by pretraining.

While QoQ-Med3 exhibits lower hallucination rate than other open source models, and demonstrates the best diagnostic performance against commercial models, we caution that the model still hallucinates some answers throughout the validation set. Mitigating intrinsic hallucinations in particular is a critical step for prospective model deployment. These results strengthen the case for exploring how changes in architecture, refinement of the task definition, adjustments in training objective, or selection of feature exclusion, masking, or aggregation during training may affect the downstream behavior of the model, particularly in the case of either amplifying or reducing perpetuated biases in demographic disparities in healthcare.

In conclusion, we developed two multimodal reasoning-based foundation models and tested them on a held-out MIMIC-IV and a private JHU PMAP dataset collected at Johns Hopkins University hospital. Results indicate large-scale multimodal clinical pretraining provides a strong foundation that enhances downstream robust task performance after domain-specific finetuning. Further, knowledge acquired through large-scale multimodal pretraining on non-EHR data is largely transferrable to EHR data analysis using our model. Our reasoning-mechanism also enhanced model interpretability and lead to a reduction in hallucination behaviour.

Methods

With our training method and unified multimodal model, we design two variants of the model: QoQ-Med3 and QoQ-Med3-MIMIC. We train QoQ-Med3 on CLIMB⁵⁵, a combination of 33 multimodal datasets spanning across 1D (ECG), 2D (Chest X-ray, Mammography, Dermoscopy, histopathology, Fundus), and 3D (Ultrasound, MRI, CT Scan) data with a total size of 2.61 samples. With QoQ-Med3, we analyze model’s trained performance handling diverse multimodal data for basic diagnosis of diseases. With QoQ-Med3-MIMIC, on the other hand, we test model’s ability to integrate highly multimodal data and long context EHR, and the ability to conduct complex multistep reasoning under minimal supervision.

Dataset

We outline the datasets used to train QoQ-Med3 and QoQ-Med3-MIMIC below.

QoQ-Med3 is trained on CLIMB⁵⁵. CLIMB contains 44 publicly available datasets across 13 domains. In this work, we use a subset of 33 datasets covering ECG, 2D imaging (Chest X-ray, Mammography, Dermoscopy, Histopathology, Fundus), and 3D imaging (Ultrasound, MRI, CT Scan), totaling 2.61 million samples. The remaining CLIMB datasets cover modalities not

Table 5. Details of individual datasets used in training of QoQ-Med3.

Dataset	# Classes	Classes
PTB-XL ⁶⁴	7	Normal, Conduction Delay (CD), Hypertrophy (HYP), Myocardial Infarction (MI), Sinus Tachycardia/Bradycardia/Conduction (STTC), Atrial Fibrillation/Atrial Flutter (A. Fib/Aflutter), Other
Chapman-Shaoxing ⁶⁵	7	Same as PTB-XL
Georgia ⁶⁶	7	Same as PTB-XL
CPSC ⁶⁷	7	Same as PTB-XL
CheXpert ⁶⁸	14	Atelectasis, Cardiomegaly, Consolidation, Edema, Enlarged Cardiomeastinum, Fracture, Lung Lesion, Lung Opacity, Pleural Effusion, Pneumonia, Pneumothorax, Pleural Other, Support Devices, No Finding
MIMIC-CXR ⁶⁹	14	Same as CheXpert
VinDr-CXR ⁷⁰	6	Lung tumor, Pneumonia, Tuberculosis, COPD, Other diseases, No finding
COVID-19 ⁷¹	4	Normal, Bacterial Pneumonia, COVID-19, Viral Pneumonia
CoronaHack ⁷²	3	Normal, Bacterial Pneumonia, Viral Pneumonia
VinDr-Mammo ⁷³	5	BI-RAD 1-5
CBIS-DDSM ⁷⁴	6	BI-RAD 0-5
CMMD ⁷⁵	2	Benign, Malignant
ISIC-2020 ⁷⁶	2	Malignant, Benign
HAM10000 ⁷⁷	5	Melanoma (MEL), Nevus (NV), Basal Cell Carcinoma (BCC), Actinic Keratosis/Intraepithelial Carcinoma (AKIEC), Other (OTHER)
PAD-UFES-20 ⁷⁸	5	Melanoma (MEL), Nevus (NV), Basal Cell Carcinoma (BCC), Actinic Keratosis/Intraepithelial Carcinoma (AKIEC), Other (OTHER)
Messidor-2 ⁷⁹	5	None, Mild DR, Moderate DR, Severe DR, PDR
APTOS 2019 ⁸⁰	5	No DR, Mild, Moderate, Severe, Proliferative DR
Jichi ⁸¹	3	SDR (simple diabetic retinopathy), PPDR (pre-proliferative diabetic retinopathy), PDR (proliferative diabetic retinopathy)
LNDb ⁸²	3	nodule $\geq 3\text{mm}$, nodule $<3\text{mm}$, non-nodule
INSPECT ⁸³	5	No PE, Acute Subsegmental-only PE, Acute PE, Subsegmental-only PE, Chronic PE
KiTS23 ⁸⁴	2	Benign, Malignant
Hemorrhage ⁸⁵	2	No Hemorrhage, Has Hemorrhage
RSPECT ⁸⁶	3	No PE, Chronic PE, Acute PE
BUSI ⁸⁷	3	Normal, Malignant, Benign
COVID-BLUES ⁸⁸	2	Has COVID, No COVID
COVID-US ⁸⁹	3	Covid, Pneumonia, Normal
Brain Tumor ⁹⁰	4	No Tumor, Pituitary Tumor, Glioma Tumor, Meningioma Tumor
Brain Tumor 2	2	Yes, No (presence of tumors)
LC25000 ⁹¹	5	Colon adenocarcinomas, Benign colon, Lung adenocarcinomas, Lung squamous cell carcinomas, Benign lung
BCSS ⁹²	4	Tumor, Stroma, Lymphocytic infiltrate, Necrosis/debris

supported by our current architecture (e.g., genetic and graph data). The full dataset list is provided in Table 5. All 33 datasets used from the CLIMB benchmark are publicly available and contain fully deidentified data. Each constituent dataset was collected and released under its own institutional ethics approval and informed consent procedures, as detailed in the respective original publications cited in Table 5. As this study exclusively uses previously published, publicly available, deidentified datasets, no additional IRB approval was required. We curate and generate the QA version of the datasets, created in the following way: For every sample x_i with label y_i , a paired question and answer is generated as follows:

Question: a natural-language query formed by combining: 1. a brief task prompt (e.g., “identify the primary abnormality”); 2. a cue about the input modality (e.g., “on the chest X-ray”); 3. the list of allowable labels drawn from the dataset.

Answer: the gold label derived from y_i , formatted as: 1. a single class a_i when categories are mutually exclusive, or 2. a comma separated subset of labels when multiple findings may co-occur.

An example QA pair is as follows:

Question. “Shown above is a frontal chest radiograph. State the predominant finding, choosing from: No Finding, Cardiomegaly, Pleural Effusion, Lung Consolidation, Atelectasis.”

Answer. “Pleural Effusion”

For multi-label tasks, predictions are judged with an order-independent match: a response is deemed correct if all ground truth labels are present in the answer, and no other labels are present, irrespective of label ordering. This criterion accommodates scenarios where several conditions may be present simultaneously and listed in any sequence. In addition, a subset of the clinical modalities presented in CLIMB are labeled as the following three categories:

1. **Novel Tasks.** Novel tasks are defined as new diseases or conditions not present previously. Examples include COVID X-ray and COVID ultrasound.

2. **Understudied Modalities.** Building on the criteria set forth in the original study, we characterize each modality along two axes: (i) *research attention*, estimated by the number of Google Scholar hits returned for standardized queries such as “[modality] classification,” and (ii) *public data availability*, counted as the total number of openly released samples. Based on both criteria, mammography and ECG appear under-studied in the literature, whereas ultrasound is chiefly limited by data scarcity.

3. **Underrepresented Regions.** The underrepresented regions are defined as datasets collected from regions historically underrepresented regions (like Africa, South America and Parts of South and Southeast Asia), or regions with less economic developments (like India, Vietnam, Iraq and Brazil). These categories demonstrates datasets with particular clinical significance, due to their limited representation in prior studies.

QoQ-Med3-MIMIC is trained on MIMIC-IV⁶³ dataset. MIMIC-IV is a deidentified set of electronic health record (EHR) data spanning over 65,000 ICU patients and 200,000 emergency department patients seen during 2008 - 2022 at the Beth Israel Deaconess Medical Center in Boston, MA. To this dataset, we link MIMIC-CXR⁶⁹, a dataset of 377,110 chest radiographs and related radiology reports from 227,835 radiographic studies, and MIMIC-IV-Note⁹³, a dataset of 331,794 deidentified discharge summaries from 145,915 patients. The MIMIC-IV dataset and its linked components (MIMIC-CXR, MIMIC-IV-Note) were collected under a protocol reviewed by the Institutional Review Board (IRB) at the Beth Israel Deaconess Medical Center (BIDMC), which granted a waiver of informed consent and approved the data sharing initiative. All data are fully deidentified in accordance with the Health Insurance Portability and Accountability Act (HIPAA) Safe Harbor provision. As this study exclusively uses publicly available, deidentified datasets accessed through PhysioNet credentialed data use agreements, no additional IRB approval was required. At a minimum, the data attributed to each patient from MIMIC-IV spans tabular information (chart values, lab work values, procedures), demographic information (deidentified age and gender), and final diagnoses in the form of ICD codes and their natural language description. 370041 (68.84%) patients’ records also contain natural language (328525 (61.12%) with discharge notes, 209007 (38.88%) with radiology report notes), and 212923 (39.60%) patients’ records contain radiographs. We curate and generate the QA version of the MIMIC-IV data in the following manner: For every sample x_i with label y_i , a paired question and answer is generated as follows:

- **Question:** a natural-language query formed by combining: 1. a summary of the patient’s information at all encounters prior to and including the current hospital admission; 2. one of five task prompts (e.g., "Provide a numbered list of 3 to 10 most likely diagnoses for this patient for their current admission."); 3. a final system prompt with the desired answer format (e.g., "The reasoning process MUST BE enclosed within <think> </think> tags. The final answer MUST BE put in boxed{ }").
- **Answer:** the label derived from y_i , formatted as: 1. a single multiple choice letter answer a_i when the task is multiple choice, or 2. a natural-language answer when the task is open-ended, stating the full list of the diagnoses associated with the hospital admission in question.

To prevent information leakage, we mask all label-revealing and outcome-encoding fields of the current admission, including its diagnosis ICD codes, discharge summary, radiology report, and discharge and death times. Only notes from prior admissions are retained as patient history; any note whose chart time falls at or after the current admission’s admit time is removed.

The five task prompts generated from the MIMIC-IV data are as follows:

1. **Open ended, list of diagnoses (T1):** "Provide a numbered list of 3 to 10 most likely diagnoses for this patient."
2. **Open ended, single diagnosis (T2):** "What is the projected diagnosis of the patient at this time?"
3. **Multiple choice, single diagnosis (T3):** "What is the projected diagnosis of the patient at this time?"
4. **Multiple choice, survival likelihood (T4):** "Will the patient survive for at least 48 hours? Answer with yes or no."
5. **Multiple choice, length of stay (T5):** "How long will the patient stay in the hospital? A. 0-4 days; B. 5-8 days; C. 9-12 days; D. more than 12 days."

A truncated example of a QA pair (T1) is as follows:

Question. "Below is a history of a patient: Patient is a 83.0-year-old F.

Procedures performed: Percutaneous vertebroplasty (ICD code: 8165); Electrocardiogram (ICD code: 8952); ...

Lab results: Anion Gap: 14.0 mEq/L (2205-03-25 07:22:00); Bicarbonate: 23.0 mEq/L (2205-03-25 07:22:00); ...

Radiology notes: The radiology note observed at chart time 2205-03-23 11:18:00 is: EXAMINATION: MRI OF THE LUMBAR SPINE INDICATION: History: low back pain, foot numbness ...

Additional medical context from other prior hospital admissions...

Provide a numbered list of 3 to 10 most likely diagnoses for this patient for their current admission.

The reasoning process MUST BE enclosed within <think> </think> tags. The final answer MUST BE put in boxed{ }

Answer. "All diagnoses from this admission: B02.8-Zoster with other complications; R07.9-Chest pain, unspecified; E03.9-Hypothyroidism, unspecified; I10-Essential (primary) hypertension; ... R68.89-Other general symptoms and signs."

We provide truncated example of each type of QA pair in Supplement. Figure 5(b), (c), and (d) presents the demographic breakdown of the MIMIC-IV QA pairs used in the finetuning pipeline.

Data from JHU includes patients with admission to inpatient and outpatient clinical centers and (1) an anesthesia data record; (2) an admission, discharge, or transfer from a location within the Department of Anesthesiology and Critical Care Medicine; (3) an encounter with the chronic pain clinic; (4) a pain service consult; (5) an evaluation in a pre-operation clinic; or (6) a difficult airway. The study was approved by the Johns Hopkins Institutional Review Board (IRB00464029). Data were collected from August 2009 through August 2024. Data include patient and encounter metadata, structured clinical variables (e.g., laboratory results, International Classification of Diseases (ICD) codes, and medication orders), and text data (e.g., electrocardiogram reports, as read by board-certified cardiologists). The dataset encompasses over 8.3 million inpatient encounters and 863,000 unique patients, with data collected from hospital information systems including demographic information, admission and discharge records, diagnoses with ICD-10 codes, medication administration, blood product administration, surgical procedures, intensive care unit (ICU) stays, and mechanical ventilation records. For every sample x_i with label y_i , a paired question and answer is generated as follows:

- **Question:** a natural-language query formed by combining: 1. a comprehensive summary of the patient's demographic information, clinical diagnoses, critical care status, severity indicators, and admission context; 2. one of six task prompts; 3. a final instruction with the desired answer format (e.g., "First, provide your answer by stating ONLY the letter (A, B, C, or D). Then on a new line, provide a brief explanation (2-3 sentences maximum)").
- **Answer:** the label derived from y_i , formatted as: a single multiple choice letter answer a_i when the task is multiple choice, corresponding to the ground truth category determined from clinical outcome data.

The six task prompts with corresponding ground truth sources and category definitions are as follows. We provide truncated example of QA pairs in Supplement.

1. **Length of stay (LOS):** "Based on the patient's admission and discharge times, which category best describes their hospital length of stay? A. Short stay (0-2 days); B. Moderate stay (3-7 days); C. Extended stay (8-14 days); D. Long-term stay (>14 days)."
2. **Ventilator duration:** "What is the total duration of mechanical ventilation (invasive ventilation only, level 6-7) for this patient during the encounter? A. No invasive ventilation; B. Less than 24 hours; C. 24-72 hours; D. Greater than 72 hours."
3. **Sepsis diagnosis:** "Did this patient receive a diagnosis of sepsis, severe sepsis, or septic shock during this encounter? A. No sepsis diagnosis; B. Sepsis (ICD-10 A40-A41, without organ dysfunction); C. Severe sepsis (ICD-10 R65.20); D. Septic shock (ICD-10 R65.21)."
4. **Bleeding severity:** "Did this patient experience a clinically significant bleeding event during this encounter? A. No documented bleeding; B. Minor bleeding not requiring transfusion; C. Moderate bleeding requiring 1-3 units RBC transfusion; D. Major bleeding requiring >3 units RBC transfusion or massive transfusion protocol."
5. **30-day mortality:** "Did this patient die within 30 days of hospital discharge from this encounter? A. Alive at 30 days post-discharge; B. Died within 30 days of discharge; C. Unknown - insufficient follow-up data; D. Still hospitalized at 30 days."

Table 6. Comparison of QoQ-Med3 against other open-source public clinical MLLMs. BBox: Ability to predict salient bounding boxes; CXR: Chest X-ray; Mammo.: Mammography; Derm: Dermoscopy; Patho.: Histopathology; US: Ultrasound. BBox: Whether the model is able to produce bounding boxes as output. QoQ-Med3 is currently the largest public medical reasoning MLLM in the field, and the only model with the ability to do cross modality reasoning with raw time series and vision inputs.

Model	Size	Training	BBox	1D	2D					3D		
				ECG	CXR	Mammo.	Derm.	Fundus	Patho.	US	MRI	CT
LLaVa-Med ⁹⁴	7B-13B	SFT	✓	✗	✓	✗	✓	✗	✓	✗	✓	✓
Med-Flamingo ¹⁴	8.3B	SFT	✗	<i>o</i> *	✓	✗	✓	✗	✓	✗	✓	✓
RadFM ⁹⁵	14B	SFT	✓	✗	✓	✓	✓	✗	✗	✓	✓	✓
BiomedGPT ¹³	33M-182M	SFT	✓	<i>o</i> *	✓	✗	✓	✓	✓	✓	✓	✓
Med-R1 ⁴⁵	2B	GRPO	✗	✗	✓	✗	✓	✓	✓	✓	✓	✓
QoQ-Med3 (Ours)	8B-32B	DRPO	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

6. **In-hospital mortality (IHM):** "Did this patient die during this hospitalization (in-hospital mortality)? A. Survived to hospital discharge; B. Died during hospitalization; C. Unknown status; D. Transferred to another facility (status uncertain)."

Model Design

We design the model with an aim that it can take in data across as many domains as possible, so that it can provide comprehensive diagnosis while correlate and co-train across the most diverse range of clinical domains, with inputs ranging from 1D to 3D. Table 6 demonstrates that our model is currently the largest open clinical MLLM in the field. It is also the only model that can both take in time series data and output its thinking process, along with the bounding box annotation highlighting the salient region made during the thinking process.

As shown in Figure 9, we initialize QoQ-Med3 from a large pretrained vision-language model comprising an image encoder, a linear projection that maps each visual patch embedding into the backbone LLM’s token space, and the LLM. To ingest temporal data, we prepend a pretrained time-series encoder, namely ECG-JEPA⁹⁶, whose outputs are passed through a newly initialized linear projection of matching dimension. At inference, the projected image patches, time-series tokens, and tokenized text are interleaved in their original temporal order and fed to the LLM. The LLM autoregressively generates a free-text chain of thought, bounding-box tokens that localize the evidence identified in that reasoning, and outputs a short diagnosis. This design supports heterogeneous modality combinations, allowing the model to skip missing channels while preserving positional consistency across the multimodal sequence. Comparing with the previous generation model QoQ-Med, QoQ-Med3 in this work features updated backbone, Qwen3-VL-8B, allowing it to reach better performance than the previous generation. We adopt ECG-JEPA as a pretrained time-series encoder for ECG representation extraction. Our framework enables integrating these representations with vision and text through the LLM backbone for cross-modal reasoning.

Training Process

Training proceeds in two stages.

Stage 1: Modality alignment. Since we initialize the projection layer from scratch, we first train and align the ECG encoder, the projection layer, and the LLM. To encourage high-quality reasoning outputs from the beginning, we use the same DRPO training as in Stage 2.

Stage 2: Multimodal fine-tuning with DRPO. We train on the full multimodal corpus with Domain-aware Relative Policy Optimization (DRPO), a reinforcement learning training method that balances training across different samples in various domains and difficulty. In this stage, we aim to simultaneously improve the diagnostic accuracy and reasoning quality. In this work, we further tune the model with downstream task with MIMIC-IV, providing detailed analysis of its performance with complex tasks requiring multi-step reasoning.

QoQ-Med3-8B trains in approximately 48 hours on 8×NVIDIA A100 GPUs with a per-device batch size of 4 and rollout batch size of 512. The 32B variant requires approximately 2 weeks on 8×H200 GPUs. Compared to PPO-based alternatives that require a separate value network of comparable size to the policy, DRPO adds negligible overhead to the standard GRPO training loop: as shown in Supplementary Table 4, DRPO requires 429 seconds per training step compared to 425 seconds for GRPO, an increase of less than 1%. The total compute for the 8B model is approximately 768 A100-GPU-hours, making the approach practical for academic research groups with standard cluster access.

Code Availability

All code, including the data processing pipeline and model training pipelines, are available publicly under [this link](#). All trained models are public and can be accessed via Huggingface directly. The list of model weights are available under the same link above.

Ethics statement.

The private JHU PMAP dataset was collected under Johns Hopkins Institutional Review Board approval (IRB00464029) with all patient data de-identified and in accordance with the Declaration of Helsinki. MIMIC-IV and CLIMB are publicly available, de-identified datasets released under data use agreements and are IRB-exempt for secondary research use. No identifiable patient information was used at any stage of model training or evaluation.

Domain-aware Relative Policy Optimization (DRPO)

Group Relative Policy Optimization (GRPO) is a reinforcement learning method that gained prominence following the success of DeepSeek-R1. Unlike Proximal Policy Optimization (PPO), which relies on a separate value network to estimate advantages, GRPO directly computes the advantage $\hat{A}_{(q,i,t)}$ for each response within a group of rollouts $G(q,t)$ at a given training iteration. A rollout refers to a single sampled trajectory or response generated by the policy in reaction to a prompt. The advantage quantifies how much better a particular rollout is compared to others generated for the same prompt, enabling the policy to prioritize relatively high-quality responses without requiring an explicit estimate of expected return. We include a summary of the algorithm in Supplement.

Reward Design

During the training of QoQ-Med3, we employ a combination of two main rewards and two auxiliary rewards that balance diagnostic accuracy with interpretability, a critical requirement for clinical applications where understanding model reasoning.

The primary goal of our model is diagnostic accuracy, for which we compute a standard *accuracy reward* r_i^{acc} . We treat prediction $\hat{y}_i$ and ground truth y_i as unordered sets of labels and assign

$$r_i^{\text{acc}} = \text{F1}(\hat{y}_i, y_i) \quad (1)$$

which directly optimizes the model’s ability to identify correct diagnoses across diverse clinical scenarios.

For clinical applications, the ability to identify and highlight relevant regions in medical imagery is crucial for building clinician trust. The *semantic alignment reward* encourages the model to correctly identify salient regions that support its diagnostic decisions. Let $\mathbf{b}_i = \{b_{i,j}\}_{j=1}^{K_i}$ be the set of axis-aligned bounding boxes output by the model and $S_i \subseteq [0, 1]^{H \times W}$ the pixel-level segmentation mask associated with the ground-truth diagnosis. We define this reward as the best intersection-over-union score:

$$r_i^{\text{IoU}} = \max_{j=1, \dots, K_i} \frac{\text{area}(b_{i,j} \cap S_i)}{\text{area}(b_{i,j} \cup S_i)}. \quad (2)$$

By optimizing this reward, the model learns to visually highlight the specific anatomical regions relevant to its diagnosis, providing critical interpretability for clinical decision support.

Accuracy offers sparse reward signals that are most useful for multiple choice questions with discrete answer sets. To better train open-ended response questions with dense reward signals, we employ a Jaccard similarity reward over the words in the model’s final answer. Let $W_{\hat{y}}$ denote the set of unique words in the predicted response $\hat{y}_i$ and W_y denote the set of unique words in the ground-truth answer y_i , both after lowercasing and removing stopwords. The similarity reward is defined as:

$$r_i^{\text{sim}} = \frac{|W_{\hat{y}} \cap W_y|}{|W_{\hat{y}} \cup W_y|} \quad (3)$$

This reward provides continuous feedback proportional to the lexical overlap between the prediction and ground truth.

The final scalar reward supplied to DRPO is a weighted combination:

$$r_i = \lambda_{\text{acc}} r_i^{\text{acc}} + \lambda_{\text{IoU}} r_i^{\text{IoU}} + \lambda_{\text{sim}} r_i^{\text{sim}} \quad (4)$$

In our experiments, we set $(\lambda_{\text{acc}}, \lambda_{\text{IoU}}, \lambda_{\text{sim}}) = (0.6, 0.2, 0.2)$.

Measures

For all experiments on multiple choice questions, we use balanced accuracy and F1 scores as performance metrics. For open-ended questions, we use BioBERT similarity as the primary measure of performance. A detailed description of metrics is included in the Supplement.

Failure Mode Analysis

When a model evaluates multiple choice tasks in a validation setting, model outputs are fully determined by its parametric knowledge. Under this condition, any incorrect model output necessarily reflects one of two outcomes: either a misjudgment of the admission to be evaluated in the patient’s context (intrinsic hallucination), or a gap in the model’s knowledge (extrinsic hallucination). We apply these two categories exhaustively to incorrect responses to the multiple choice tasks in order to characterize failure modes.

Starting with our validation dataset, we filter out open ended tasks and objectively correct model outputs, resulting in a total of 2,495 incorrect, multiple choice, validation items across all four models. To investigate the failure modes of QoQ-Med-MIMIC against the other three model variants, we utilize GPT-5.5 to perform downstream hallucination analysis on all 2,495 items, and randomly select 100 samples encompassing all model variants for two human annotators to evaluate on the same criteria.

The criteria is defined as follows:

(i) Intrinsic hallucination: The model answered using or referencing the wrong admission in the patient’s history — e.g. it grounded its answer in a prior admission or unrelated history instead of the current admission. The error is in admission identity.

(ii) Extrinsic hallucination: The model correctly identified the current admission and its details, but drew an incorrect clinical conclusion from them, or fabricated a detail not present in the current admission. The error is in the reasoning.

To encourage improved accuracy by the LLM-as-a-judge, the prompt includes instructions to perform 2-3 sentences worth of reasoning before producing a final one-word label. The full system prompt is included in the Supplement.

Data Availability

This work utilizes two public datasets, CLIMB⁵⁵ and MIMIC-IV⁶³, and a private dataset, JHU ACCM-PMAP. The public datasets are available to download from their corresponding websites, given proper permissions have been granted.

Author Contribution

D.D., J.S. conducted experiments and prepared figures for each experiment. J.C. and P.L. wrote Section 1 and Abstract. X.H., C.H., S.S. and R.S. prepared the JHU PMAP dataset, conducted experiments on it and wrote Sec. 2.5. H.W. and F.V. annotated the reasoning traces as demonstrated in Figure 1 and Supplement. D.D. and J.S. wrote the remaining part of the paper not mentioned above.

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Competing interests

The authors declare no competing financial or non-financial interests.

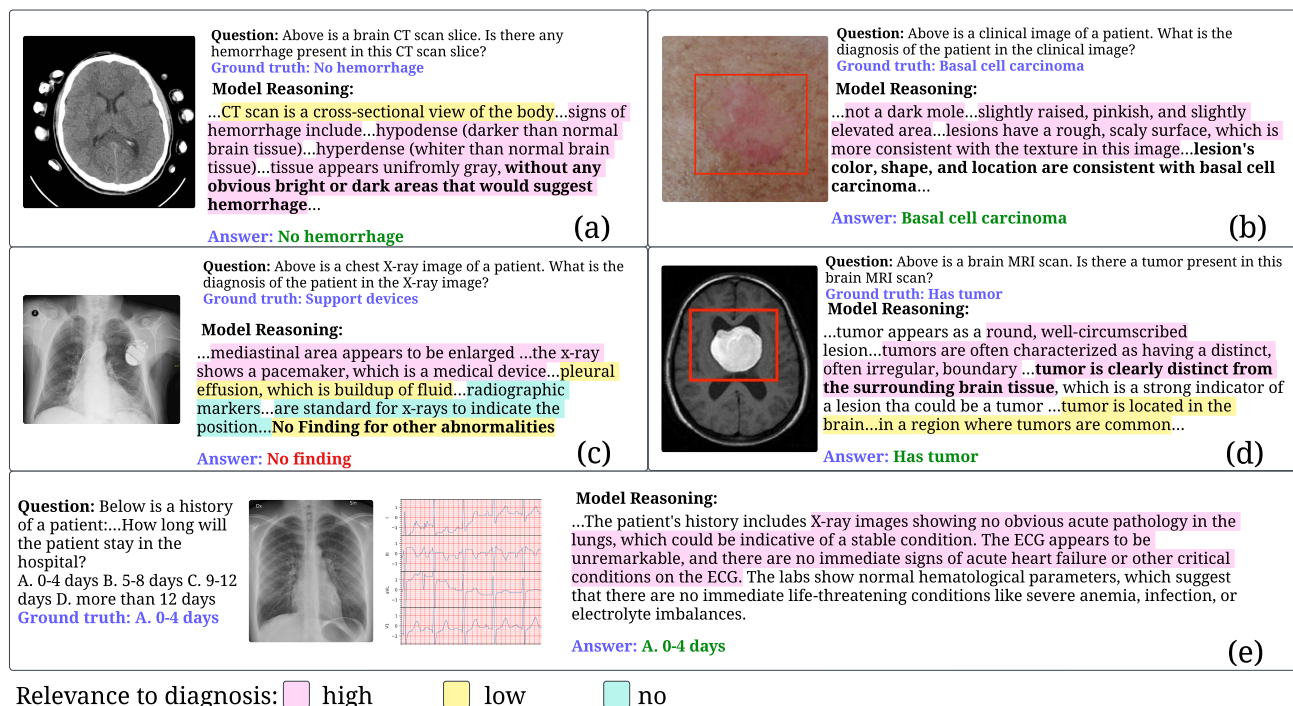


Figure 1. Model outputs annotated by clinical experts. QoQ-Med3 correctly reasons from modality-specific clinical knowledge, generates bounding boxes, and outputs the correct predictions in most instances except (c), where the model identifies a pacemaker but fails to integrate this observation into its final prediction. (e) demonstrates the model's ability to synthesize multimodal inputs including chest X-ray, ECG, and laboratory values with coherent reasoning. The bounding boxes correctly highlight the salient regions related to the reasoning steps when one is present. Color coding indicates relevance to diagnosis: high (pink), low (yellow), and none (aqua). Sample medical images are sourced from the following publicly available datasets: Brain CT Hemorrhage Dataset (CC BY 3.0, <https://www.kaggle.com/datasets/abdulkader90/brain-ct-hemorrhage-dataset>); HAM10000 Dataset (CC BY-NC-SA 4.0, <https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>); Brain Tumor Dataset (CC BY 4.0, https://figshare.com/articles/dataset/brain_tumor_dataset/1512427); MIMIC-IV-ECG (Open Data Commons Open Database License v1.0). Chest X-ray images are from Wikimedia Commons: "Normal posteroanterior (PA) chest radiograph" by Stillwaterising, available under CC0 1.0 ([https://commons.wikimedia.org/wiki/File:Normal_posteroanterior_\(PA\)_chest_radiograph_\(X-ray\).jpg](https://commons.wikimedia.org/wiki/File:Normal_posteroanterior_(PA)_chest_radiograph_(X-ray).jpg)); "Medical X-Ray imaging IHL05" by Nevit Dilmen, available under CC BY-SA 3.0 (https://commons.wikimedia.org/wiki/File:Medical_X-Ray_imaging_IHL05_nevit.jpg). All images are fully de-identified. This figure was created by the authors using Lucidchart (<https://www.lucidchart.com/pages>).

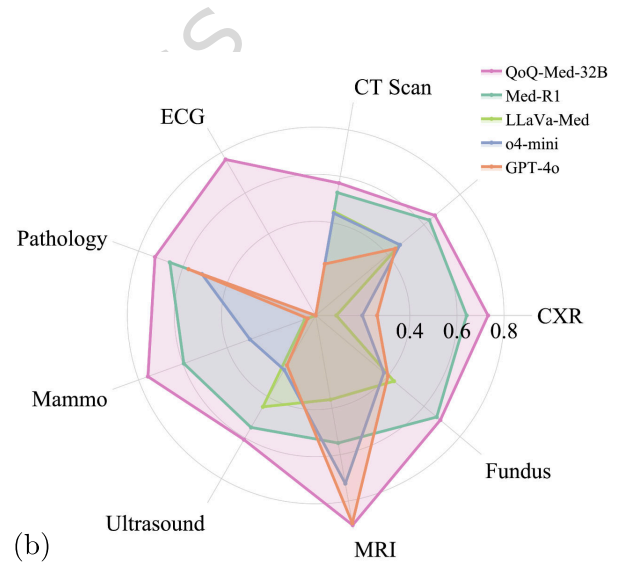
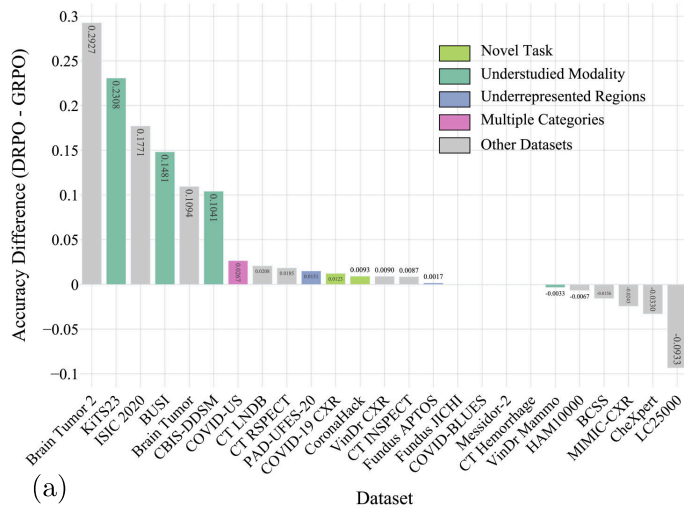


Figure 2. (a) Difference in accuracy (DRPO - GRPO). DRPO, our reinforcement learning algorithm for training on highly heterogeneous data, brings the most performance gain in understudied modalities as defined in Method Section. **(b) Accuracy comparison of QoQ-Med3 against SoTA open source and closed source LLMs.** QoQ-Med outperforms all open and closed MLLMs across 8 domains.

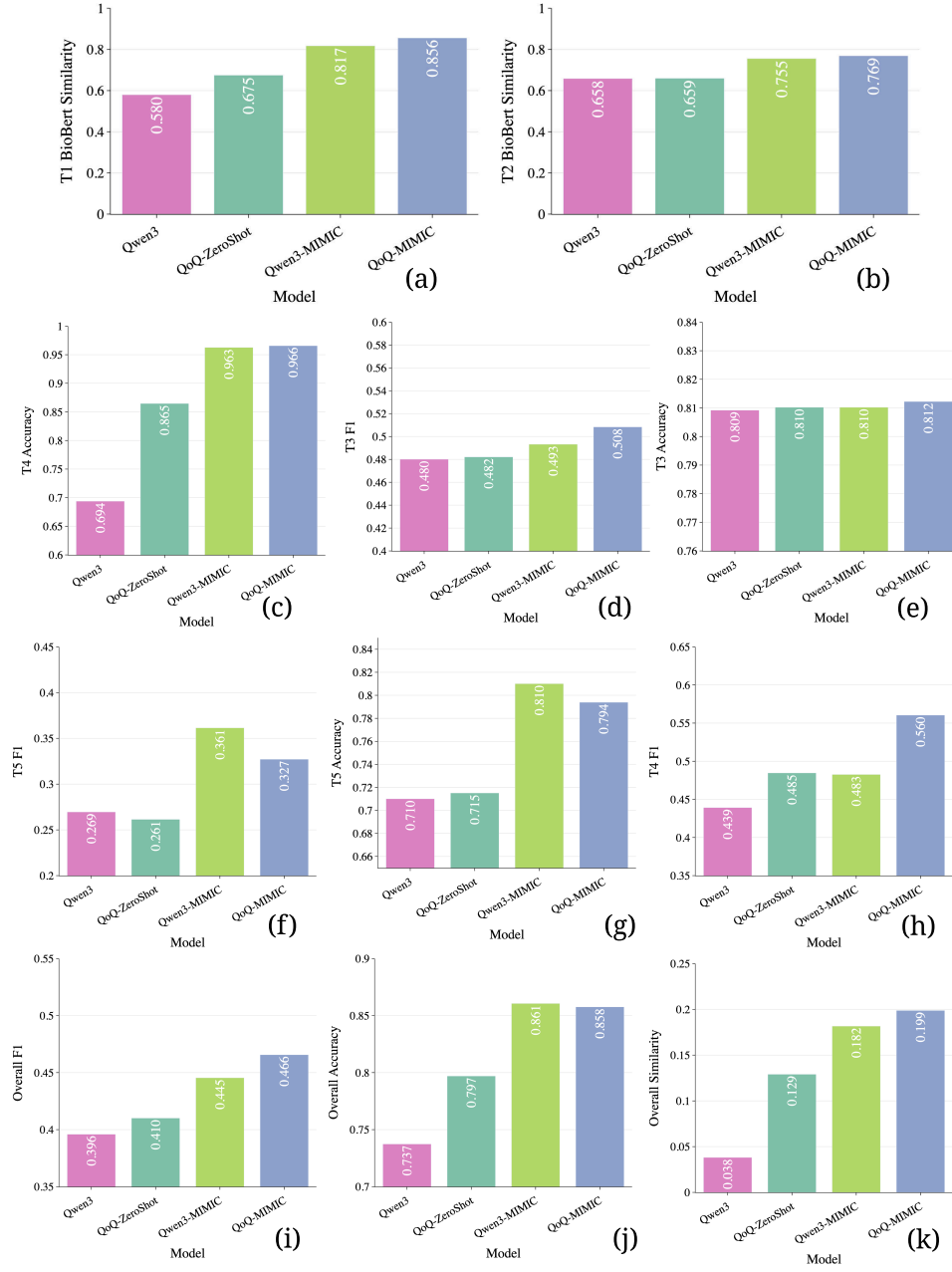


Figure 3. Performance comparison of QoQ-Med3-MIMIC and ablations on MIMIC benchmark across different question types. Qwen3 is the vanilla Qwen3-VL-8B model not trained on clinical specific data. QoQ-ZeroShot is trained on CLIMB, which includes ECG, vision, and text data, but has no exposure to any MIMIC-IV data. Qwen3-MIMIC is the vanilla model directly tuned on MIMIC-IV. QoQ-MIMIC is QoQ-Med3-MIMIC that is first trained on CLIMB, then tuned on MIMIC-IV. T1-T2 are open response questions evaluated by BioBERT similarity. T3-T5 are multiple choice questions evaluated by Accuracy and F1 Score. Definitions of each question type is included in Dataset Section. Sim: BioBERT Similarity, Acc: Accuracy, F1: F1 Score. In general, we observe that the model first trained on CLIMB, then tuned on MIMIC-IV reaches the best performance overall.

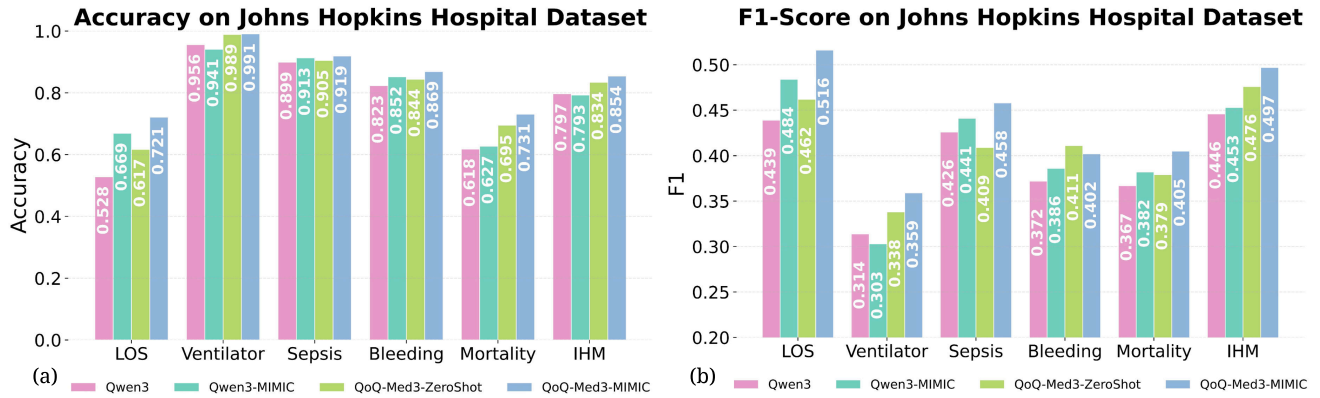


Figure 4. Performance comparison on Johns Hopkins Hospital dataset across different question types. We designed six types of multiple-choice questions that cover the most essential clinical questions arising in emergency room and inpatient settings. Detailed descriptions of each question type are provided in Dataset Section.

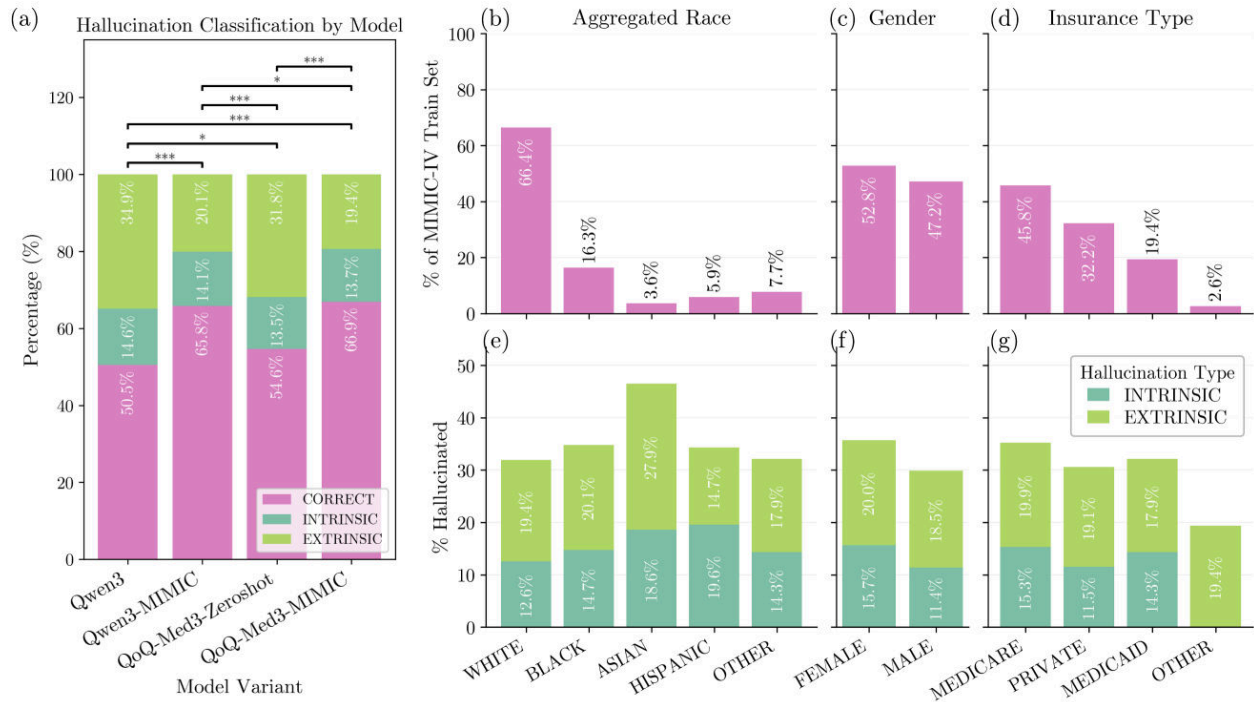


Figure 5. Hallucination classification breakdown and demographic subgroups present in MIMIC-IV training dataset. (a) Categorized hallucinations for each model variant across all multiple choice questions in the validation set ($n = 1539$ per model variant), using GPT-5.5 as the judge. Significance bars are labeled with stars corresponding to the p -value from the χ^2 -test of independence, assessing whether hallucination behavior differed across model variants. *** indicates p -value < 0.001 , ** indicates p -value < 0.01 , and * indicates p -value ≥ 0.01 . The MIMIC-IV Q&A pair training dataset ($n = 423,507$) is categorized according to demographic breakdowns, including: (b) aggregated race, (c) gender, and (d) insurance type. Hallucination rates for QoQ-Med3-MIMIC outputs in the validation set are categorized according to the same demographic breakdowns: (e) aggregated race, (f) gender, and (g) insurance type.

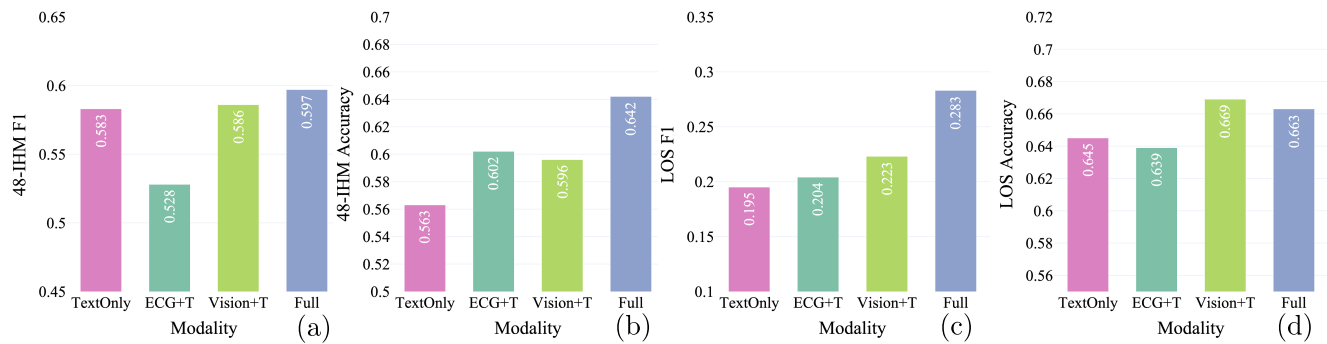


Figure 6. QoQ-Med3's Performance with partial modalities on MIMIC-IV. (a) F1 performance of 48 hour in hospital mortality (48-IHM) classification. (b) Accuracy performance of 48 hour in hospital mortality (48-IHM) classification. (c) F1 performance of 4-class length of stay (LOS) classification. (d) Accuracy performance of 4-class length of stay (LOS) classification. The full variant contains inputs from 3 modalities has the best performance for both tasks, time-series only (TS+T) and vision only (Vision+T) ablations having worse performance, and the text only ablation having the worst performance.

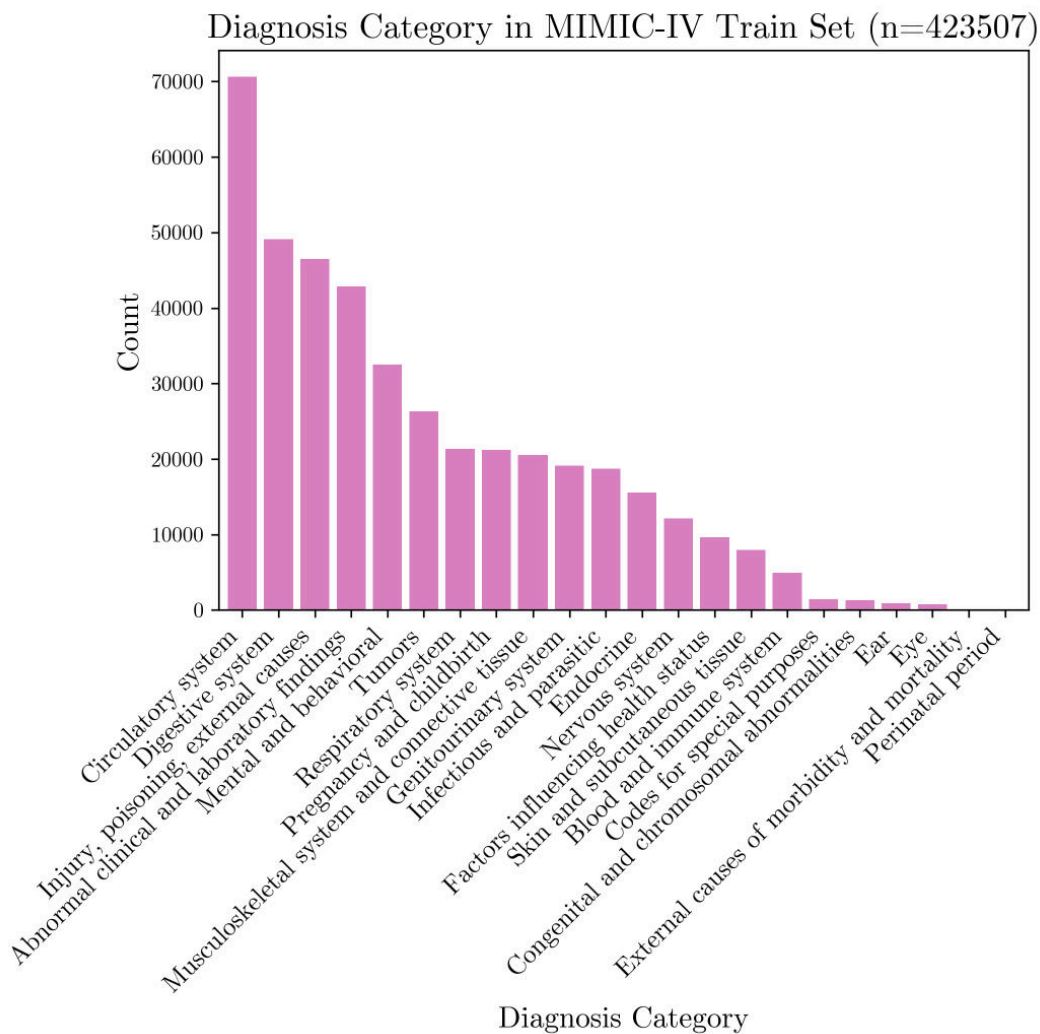


Figure 7. MIMIC-IV training dataset broken down by primary diagnosis category. The MIMIC-IV Q&A pair training dataset (n=423,507) is broken down according to the primary diagnosis category of the hospital admission.

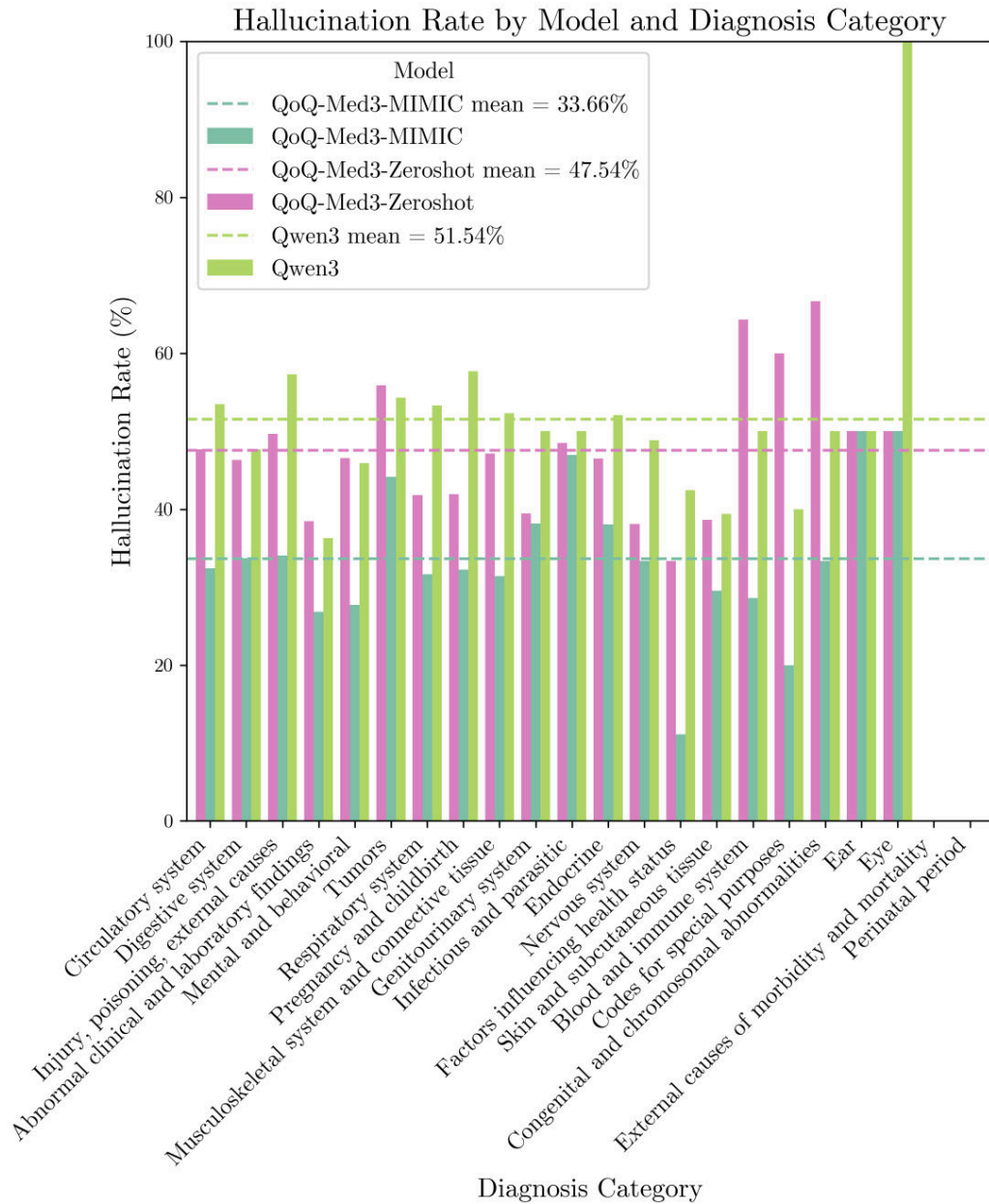


Figure 8. Hallucination rates on MIMIC-IV evaluation dataset broken down by primary diagnosis category.

Hallucination rates for Qwen3-VL, QoQ-Med3-Zeroshot, and QoQ-Med3-MIMIC outputs ($n=2,565$ for each model variant) are broken down by primary diagnosis category, in order of prevalence in MIMIC-IV training dataset. GPT-5.5 served as the judge.

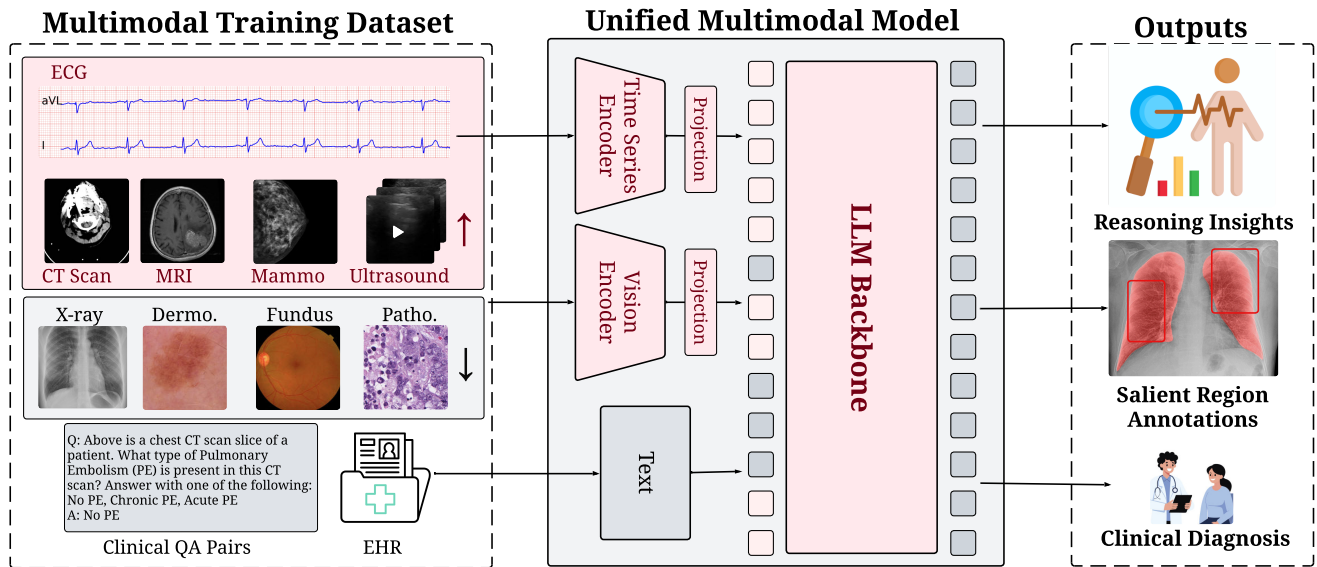


Figure 9. Overview of QoQ-Med3. The training corpus spans 11 clinical domains, including structured waveforms (e.g., ECG), diverse imaging modalities, electronic health records, and curated clinical QA pairs. Modality-specific encoders convert inputs into token embeddings that are linearly projected into a common space and interleaved with text tokens before entering the LLM backbone. The model then autoregressively produces (i) an explainable chain-of-thought, (ii) bounding-box annotations highlighting salient regions, and (iii) a concise clinical diagnosis. Sample medical images are sourced from the following publicly available datasets: Brain CT Hemorrhage Dataset (CC BY 3.0, <https://www.kaggle.com/datasets/abdulkader90/brain-ct-hemorrhage-dataset>); HAM10000 Dataset (CC BY-NC-SA 4.0, <https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>); Brain Tumor Dataset (CC BY 4.0, https://figshare.com/articles/dataset/brain_tumor_dataset/1512427); MIMIC-IV-ECG (Open Data Commons Open Database License v1.0); COVID-US Dataset (GNU AGPL 3.0, <https://github.com/nrc-cnrc/COVID-US>); CMMD (CC BY 4.0, <https://www.cancerimagingarchive.net/collection/cmmd/>); Messidor-2 (CC0 1.0, <https://www.kaggle.com/datasets/google-brain/messidor2-dr-grades>); LNDb Dataset (CC BY 4.0, <https://zenodo.org/records/8348419>); ISIC-2020 (CC BY-NC 4.0, <https://challenge2020.isic-archive.com/>). Chest X-ray image from Wikimedia Commons: “Normal posteroanterior (PA) chest radiograph” by Stillwaterising, available under CC0 1.0 ([https://commons.wikimedia.org/wiki/File:Normal_posteroanterior_\(PA\)_chest_radiograph_\(X-ray\).jpg](https://commons.wikimedia.org/wiki/File:Normal_posteroanterior_(PA)_chest_radiograph_(X-ray).jpg)). Some illustrative graphics in this figure were generated using Google Gemini by the authors. This figure was created by the authors using Lucidchart (<https://www.lucidchart.com/pages>). All images are fully de-identified.

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